Continuous extensions to the closure
Munkres, 18.13

Theorem 1. *Let $A \subset X$, let $f\colon A \to Y$ be continuous, and let Y be Hausdorff. Show that if f extends to a continuous function $g\colon \overline{A} \to Y$, then g is uniquely determined by f.*

Proof. Suppose that $g, h\colon \overline{A} \to Y$ are two continuous extensions of f, so that

$$g(a) = f(a) \qquad \text{and} \qquad h(a) = f(a)$$

for every $a \in A$, where a on the left-hand sides is regarded as a point of $\overline{A}$. We show that $g = h$.

Consider the map $A \to \overline{A}$ sending a to the same point viewed as an element of $\overline{A}$, and let R be its range. Thus R is A regarded as a subset of $\overline{A}$.

We first claim that R is dense in $\overline{A}$. Let $x \in \overline{A}$. To show that x lies in the closure of R in $\overline{A}$, let O be an open subset of $\overline{A}$ containing x. By the definition of the subspace topology, there is an open set $U \subset X$ such that

$$O = \{z \in \overline{A} \mid z \in U\}.$$

Since $x \in O$, the point x, viewed as a point of X, lies in U. Since $x \in \overline{A}$ in X, the open set U must meet A; hence there exists $y \in U \cap A$.

Since $A \subset \overline{A}$, this y is also a point of $\overline{A}$, and indeed $y \in R$. Moreover $y \in O$ because $y \in U$. Thus every open subset of $\overline{A}$ containing x meets R, so x belongs to the closure of R in $\overline{A}$. Since x was arbitrary, R is dense in $\overline{A}$.

Now let

$$E = \{x \in \overline{A} \mid g(x) = h(x)\}.$$

The set R is contained in E: if $x \in R$ comes from $a \in A$, then

$$g(x) = f(a) = h(x).$$

Since g and h are continuous and Y is Hausdorff, E is closed in $\overline{A}$. Therefore, by minimality of the closure, the closure of R in $\overline{A}$ is contained in E. But R is dense in $\overline{A}$, so every $x \in \overline{A}$ lies in the closure of R, and hence $x \in E$. That is, $g(x) = h(x)$ for every $x \in \overline{A}$.

Thus $g = h$, so any two continuous extensions of f to $\overline{A}$ coincide. Therefore the extension g, if it exists, is uniquely determined by f. □